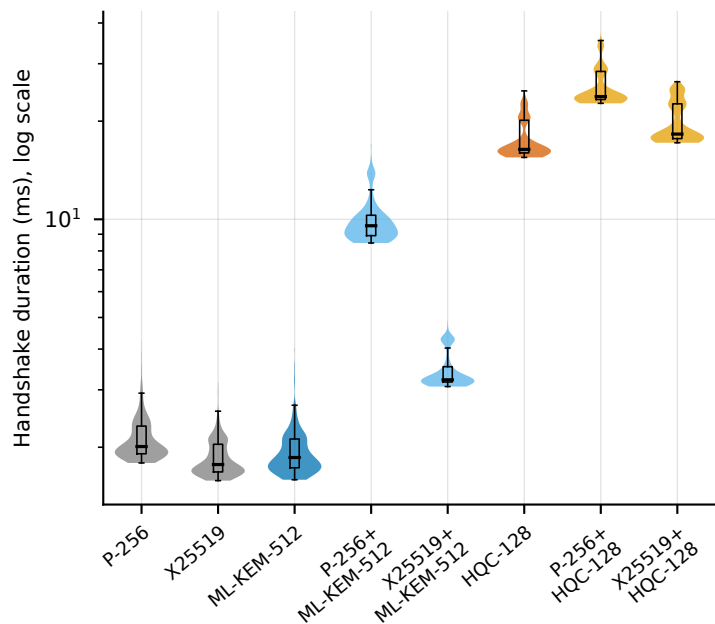
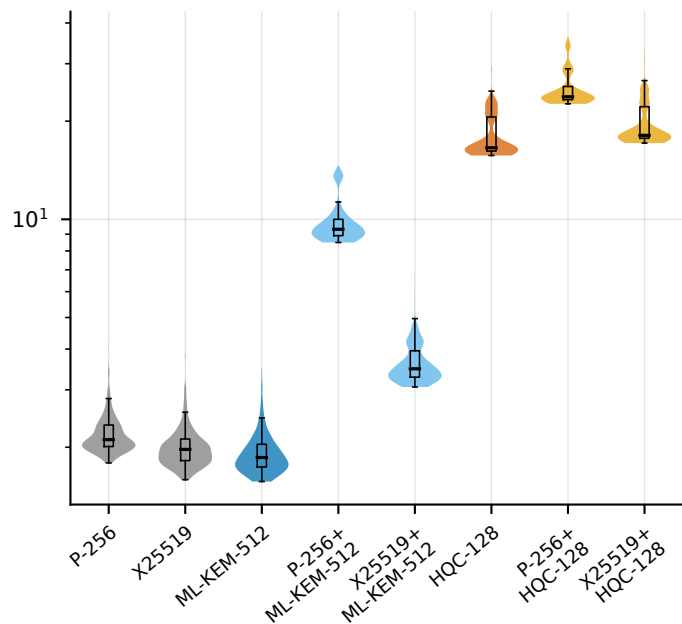


(a)

ed25519 (classical)

**(b)**

mlDSA44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mlDSA44 Mean (ms)	mlDSA44 CV (%)	mlDSA44 Max (ms)
P-256	2.15	16.1	4.22	2.22	13.7	4.18
X25519	1.87	13.2	3.16	2.02	14.4	3.87
ML-KEM-512	1.95	16.1	4.03	1.94	15.2	3.50
P-256+ML-KEM-512	10.0	16.0	17.0	9.86	15.4	14.6
X25519+ML-KEM-512	3.46	12.8	4.88	3.64	13.8	6.74
HQC-128	17.6	13.5	24.8	18.1	15.0	29.6
P-256+HQC-128	25.6	13.9	47.0	25.3	12.9	36.1
X25519+HQC-128	20.1	14.6	26.4	19.6	15.0	33.5